

Gate E2 report — run_v1.2_r1 (Paper IV v1.2)

Verdict: **INCONCLUSIVE**. Evidence directory: 20_EVIDENCE/E2/.

E2 — Independent mathematics (first pass)

Method: pen-and-paper rederivation by the lead auditor from the EN manuscript (no author derivation notes, internal reports or editorial reports were read); directed inspection of Lean sources only where the prose is a summary (recorded). Exact numerical sub-steps are cross-checked in E5 (tests T01–T24). Status legend: **R** = rederived by the auditor; **R*** = rederived given an explicitly stated intermediate estimate; **I** = imported with attribution (not re-proved); **F** = only formally inspected (Lean), no human derivation available/attempted; **X** = exposition gap (see A.2 table).

Component ledger

#	Component (EN lines)	Status	Notes
1	Loss budget (1.3), (1.3a), Lemma 2.1, (2.4), Table 1	R	pure counting
2	Finite LP duality / attained rational optimum (§2.2)	I/F	Paper I; <code>PaperI.FiniteLPDuality</code> (not re-proved)
3	(2.5) Paper II bound $\lfloor (2n+1)^2/24 \rfloor = M(n)$	R (floor identity), I (Paper II theorem)	
4	Lemma 3.2 bounded-rank nibble	I	Paper III, attributed
5	Lemma 3.3 two quotas (1.1459-1483)	R	loads/codegrees of the auxiliary extension, projection, both quota constants $C = C_0 + 1 + k_0 + b(2 + 2k_0)$ rechecked
6	Lemma 3.4 physical selector (1.1485-1504)	R	pair mass $\geq (t_3 - 3)/2$, load non-increase, codegree $\leq \gamma_0/2 + 3/t_3$ (actual cross term $\leq 2/t_3$, gain $4m + b_4$, (3.6)
7	(3.7) algebra; (3.9)	R	
8	(3.10) charging ($\times 3$ per discarded edge; K4 with one bad edge keeps a face; light K4 $\rightarrow$ faces at $x/2$)	R	discard count $(3\delta + 1/k_0 + d)n^2$ is standard regularity bookkeeping, not derived (A2-2)
9	(3.12)-(3.13) joint codegree	R*	given $\psi_H \leq t^2$ and fibre lower bounds $b_H \geq a_3 t^3, a_4 t^4$ (A2-3)
10	Per-edge load of the cleaned system (1.1615 "A single edge is controlled in the same way")	X / F	Following the stated recipe gives load $\leq 1/a_3 \gg 1$, not ≤ 1. The Lean development instead removes roots whose copy count deviates two-sidedly from the reference value <code>(RC01DeviationCleanup.deviationBad</code> , Chebyshev second-moment bound) and uses the resulting <i>upper</i> spread bound with budget $(1+u) \cdot A \cdot \text{pairVolume}$ (<code>RC01CleanFiber</code> docstring). This essential step is not stated in the prose. See A2-3.
11	C.3 schedule (C.1), (C.2), Table C.1, (C.3)	F	numeric domination with $2^{\{59516\}}$ -scale quantities; definitions of $L_T, M_T, \gamma_T, T_T, D_S, \gamma_R$ are not in the manuscript (A2-4)
12	Corollary 3.5	R	uses $M(n) \geq n^2/6$ for $n \geq 6$
13	Lemma 4.1 admissible mixed path	R	transport (4.4) (loads ≤ 2 before averaging, ≤ 1 after), existence of a non-twin step via Dirac + rooted Dirac, lexicographic potential, chordality preservation
14	Lemma 4.2, (4.5)-(4.6a), descent, near-threshold $4 \cdot 10^{12}$ / failure at $3.6 \cdot 10^{12}$	R	no circularity: Lemma 4.2 uses Thm 5.0, whose proof does not use Prop 4.3
15	Prop 4.3 (4.7)-(4.8)	R	from $ \Delta F_4 \leq 6$ per edit (actually ≤ 4) and the first comparator branch
16	(5.1a) three comparator branches	R (values of the branch maxima, e.g. $(4n+1)^2/120$), F (existence of the explicit fractional packings of the middle/third branches)	
17	Lemma 5.1 extraction	R	PEO edge bound $(a-1)r - C(a,2)$; $u \leq \sqrt{(2\varepsilon_0)n}$; $\delta_C \leq 6.1\varepsilon_0 n^2$; $a \geq 0.33n$; budget $1.6617e-6 n^2$
18	Prop 5.2 regularization, (5.2)-(5.4), (5.7)	R	incl. $ X \leq a/16384$, $ Y \leq a/57344$, exterior cliques $\leq a/128 + 1$, Y clique, R clique; $d_{\max}$: removed X-columns miss $\geq a/4$ of $\leq 129a/64$ exterior vertices $\Rightarrow \leq 113a/64$ (the source of "113" is not stated in the prose); all (5.4) ratios and (5.7) with margin
19	Phase I: equitable Vizing colouring, (5.5), PEO orientation, invalid incidences $\leq (w-1)A$, cyclic averaging (5.6)	R	

20	Phase II: (5.8)-(5.11) moments and averaging; compatibility with phase I	R	$(a+u)^2 \leq Da + 4ta + 2tu$
21	(5.12)-(5.14) calibration; Table 2; Lemma 5.3 (5.15), (5.15a); (5.16)	R	coefficients recomputed exactly (also E5 T03/T04)
22	Corollary 5.4	R*	given (4.8)
23	Theorem B assembly; §6.1 weight argument; (6.2); §6.2 (6.3)-(6.4); §6.3 Thm A with $b=N^2$	R	tower value $T(h+7)$ is F (item 11)
24	Quadratic/linear coefficient optimality (6.5a), (6.5)	R	
25	§6.4 (6.6) piece identity	R	
26	Thm 6.1, (6.7a)-(6.7b), Cor 6.1a (48, 480), Cor 6.2	R	$48 = \text{best from (6.7) (E5 T22); } \leq 10\text{-d_R bound (E5 T08)}$
27	Thm C': deletion step (6.15), $\delta' \geq 0$, $\delta' \leq \delta - \epsilon t + 1/3$, reinsertion $\leq t-1$ edits paid by $A_s \geq 2/\epsilon_s$; window invariant (F.1 l.2141-2146); repair bound $B + 4d_E$; (6.16)	R	invariant algebra also E5 T19
28	Thm C' terminal accounts (6.14) / E.4 combination	R	coefficient $1 + 32s + 800(3 + 32s)(s+1)^2 \leq 25632(s+1)^3 \leq 38000(s+1)^3$; total $\leq 40000(s+1)^3$
29	Cor 6.3 resize, (6.19) discrete parabola, (6.18)	R	parabola: $Q_s - B = [(N-3k)(N-3k+1) - r_N]/6$, $N=n+s$, $r_N \in \{0,2\}$; $\geq \text{dist}^2$ (E5 T09)
30	Prop 6.3a (6.20)-(6.21)	R	$\text{rsd} \leq s$ of $G_{s,q,d}$ (hosts: omit $ D \cap U $; C, D vertices: host-clique argument), baseline deficit $(3d^2 - d)/2$ via increments $-1-3j$, edge-count difference $d(4q-4s-d-1)/2$, $4\Delta e - nd = d(5q-7s-2d-2) \geq 0$
31	Prop 6.4 (6.22)/(6.23)	R* (low-star branch: exact identity $(n-2c+1)c + C(c,2) = B_{\{n-s\}}(c) + sc$; (F.3)); I ([15, Lemma 3.1] signed-star estimate (F.2), attributed); F (high-star envelope (F.4), small orders $n < 8$ in Lean)	independent finite check E5 T15
32	Thm 6.5	R*	given SublinearEdit (from [22] via E34, I/F), Lemma F.3a (R), (F.6) (R* — my reconstruction gives $16\delta + 65t + n(s+u)$ and $\delta + 4t + 2n(s+u)$), RootScore minimization (R)
33	Lemma F.3a, Cor F.3b	R	both induction cases; also E5 T12
34	Cor 6.6, (C.4)	F	tower comparisons (E35) not rederived
35	Appendix E.1 constructor: (E.2), (E.2a), (E.2c), (E.2e)	R	(E.2e) increments $\lfloor (N+s+1)/3 \rfloor \geq \lceil (c+b+s)/3 \rceil$
36	(E.1) capacity conditions, list hosting of all edges of $G[S]$ (Galvin, dyadic)	F	only described
37	(E.2b) three-case budget	R*	cases 1–3 arithmetic checked given (E.2d) and the normalization estimates
38	(E.2d) joint defect bound	X	sketch only (A2-6)
39	Normalization estimates (t_W , L , $ A $, $ c-n/3 $, σ , τ , m)	X / F	asserted, not derived (A2-5)
40	E.2 min-degree reduction removing K_s ; (E.3)	R	(E.3) holds for all $N \geq 1$
41	E.3 lower witness, (E.5)	R	
42	E.4 block by block: StrictParams (κ , q , capacity conditions)	R* (given normalization numbers)	$\kappa \geq 2n/3 - 0.021\rho$; light/heavy host degree bounds $\leq \kappa$; $3\lceil c/2 \rceil + 2\tau + 4q \leq b+2$
43	E.4 (E.7) θ_s -saving from cyclic averaging	R*	each H-edge has $\geq c-2\sigma$ valid hosts; $z \geq dm/\kappa \geq \theta_s m$
44	E.4 (E.8)-(E.9)	R	exact substitution of R_* and F; $D_0 \geq 0$ from (E.2e)
45	E.4 StrictBudget (E.10)	R*	depends on (E.2d) + normalization (items 38–39)
46	E.4 ResidualExcess (E.11)	R	needs $2(L+t_W) \leq (c+b+s)/3$
47	E.4 missing-pair restriction (no_core_pairs)	R*	given $ U \geq \rho/10^4 + s + 2$ from normalization
48	E.4 CoreCliqueAlternative (E.12), (E.13)	R	induction on s ; $(4s+1)s$ missing-edge bound ($4s-1$ suffices); $c \geq 20(s+1)^2$
49	E.4 TemplateDistance / RootPartition (E.14)	R	two successive comparisons

50	F.5 case structure A/B1/B2	R (assembly logic, exhaustiveness, (F.12) with $\delta_{11} \leq \varepsilon/2$, $K\eta = \varepsilon/2$, prefix-extension counting) ; I/F (adapted set-colouring theorem thm2, claim5_count, gluing lemma — from [22] with modifications)	matches Lean lemma11_V
51	(F.7)-(F.10) auxiliary lemmas	F (+ E5 finite tests for F.7, F.9)	
52	Prop D.1, Table 8, (D.1)-(D.4)	R (counts, residue coverage), I (Bose/Skolem STS existence)	
53	Prop D.2, D.3 (D.6)-(D.7), D.3 (D.8)	R	
54	G.1 annex (G.1)	F (statement only)	
55	Prop G.1 (G.2)-(G.5)	R	Behrend via Mathlib is I
56	§8 (8.1)	R	

No mathematical error was found in any rederived step.

Appendix A.2 exposition verdicts (mandate §5.2)

Item	Manuscript location	Literal Lean declarations (located)	Independent derivation attempted	Evidence	Exposition verdict	Explanation
A2-1	§5.1 (5.3), (5.4), (5.7) l.484-514	Regularization, RegularizationBounds, RegularizedRootGoal, RootRegularizationBridge	Yes, complete (component 18)	this record; E5 T06	ACCEPTABLE_SUMMARY	Every bound follows from (5.1), (5.3) and the displayed degree/width counts by rational comparison with wide margins; the one non-obvious input (removed X-columns miss $\geq a/4$ exterior vertices, giving 113a/64) should be one sentence.
A2-2	C.2 (3.10), (3.8) l.1572-1581, l.1538-1544	E17.Cleanup, E17.FarBudget, RC01RootwiseCleanupBudget, RC01RootwiseRetention	Partial: charging argument rederived; discard count reconstructed from standard regularity bookkeeping (exceptional class, irregular pairs, intra-class, sparse pairs)	component 8	ACCEPTABLE_SUMMARY for the discard counts (3.10); retention (3.8) is covered under A2-3	A referee familiar with regularity can reconstruct (3.10); the conventions (8/8 in C.3, class-size t) are stated.
A2-3	§3.2, C.2 fibre bounds $b_H \geq a_3t^3, a_4t^4$, (3.8); per-edge feasibility l.1606-1615	RC01CleanFiber, RC01DeviationCleanup, RC01RegularVolume, RC01PatternMassScale, RC01CleanedGate, RC01CleanedMixedCodegree	Yes; the codegree bound (3.13) follows; the per-edge load bound does not follow from the text	component 10	REQUIRES_EXPANSION	The feasibility (load ≤ 1) of the cleaned system needs a two-sided deviation cleanup of roots with a second-moment (Chebyshev) bound and an upper spread bound; the prose instead says "a single edge is controlled in the same way", which with the stated bounds yields only $1/a_3$. The retention (3.8) likewise depends on that cleanup and the counting lemma. This is an essential step of Theorem 3.1 absent from the human-readable proof (the formal proof contains it).

A2-4	C.3 Table C.1, (C.1)-(C.3)	E19.ScheduleBounds·E19.ChainBounds·E19.GateBounds·E18.Numeric·E17.ExplicitFarAssembly	No (definitions of the schedule functions are not given in the text)	—	REQUIRES_EXPANSION (minor impact)	Affects only the explicit tower values $T(h+7)$, $T(h+8)$ and $(C.4)/(6.28)$, not the existence of thresholds. A human referee cannot check Table C.1 from the manuscript; either give the schedule definitions or label these values as formally verified numerics only.
A2-5	E.1 normalization l.1853-1863; E.4 l.1993-2003	A4S1.IndepAll* (AllInput.params.strict, normalization producers), E33 localization	Arithmetic downstream checked (items 42–47); the normalization estimates themselves could not be derived from the text	components 39, 42–47	REQUIRES_EXPANSION	These estimates ($t_W \leq 4\rho/10^{27}$, $L \leq \rho/10^{13}+1$, $ A -n/3 \leq c+4\lambda/10^{31}$, $ c-n/3 $, σ , τ , $m \leq 2\rho^2/10^{31}$) carry the whole fixed-defect constructor, i.e. the proof of Theorem C for $s>0$ (the paper's order-four strengthening relative to [15]) and Theorem C'. They are asserted, with the localization input itself only named.
A2-6	E.1 (E.2b), (E.2d), three-case budget l.1837-1929	IndepAllPeel·IndepAllObstr·IndepAllBudget·IndepAllMain	Cases 1–3 checked given (E.2d); (E.2d) itself only sketched	components 37–38	REQUIRES_EXPANSION	(E.2d) is the key place where rooted defect is used jointly on up to $2s+2$ exceptions; the text gives an outline ("the rooted condition, applied to the induced subgraphs determined by these pairs ... bounds the joint contribution") that a referee cannot turn into a proof without substantial reconstruction.
A2-7	D.1 Table 8 l.1721-1744	HalvingFrames·ThreeRegime·CompleteStateAllOrders·exists_packing·complete_state_closes	Yes: residue coverage and gain counts (D.1)-(D.4) rederived; STS existence classical	component 52	ACCEPTABLE_SUMMARY	Classical Bose/Skolem Steiner triple systems with a standard parallel-class K4 augmentation; Proposition D.1 is a supplement not used by Theorems A–C.

§5.1 obligations (v1.2-specific) — first-pass results

- **E.4 block by block:** StrictParams R^* ; StrictBudget R^* (conditional on A2-5/A2-6 items); ResidualExcess R ; CoreCliqueAlternative R ; LargeCliqueCore / no_core_pairs R^* ; TemplateDistance R ; RootPartition R . Same witnesses: all accounts refer to one physical partition Q and one normalized triple (S,H,W) ; (E.9) uses the constructed Q , not an optimizer. Fixed- s thresholds and windows: $\gamma_s, N_s^{\text{stab}}$ match (F.1)/(6.11) term by term (E3).
- **F.3a:** exists_clique_additive_defect — hypotheses $\text{rsd} \leq s$ on G , arbitrary $D \subseteq V$, output clique $C \subseteq D$ of G with $(|D| - |C| - s)_+^2 \leq \text{ordNE}(G,D)$: genuine clique recovery (not an induced-subgraph or template assertion). R .
- **F.5:** l11_caseA / B1 / B2 conditions, exhaustiveness and common conclusion checked (E3 static; component 50). Adaptation vs [22]: sequences with replacement, $K = \text{nodes}$, enlarged pinned class, constants m_{11} , B , δ_{11} ; accurate per E7 lead verification.
- **Theorem C' quantifier:** one root (C,D,H) chosen before every Q and τ (Lean $\exists CDH \dots \forall Q \tau$). Linear bound is to an actual-root template of arbitrary core size; optimal-size comparison carries $n\sqrt{(1+4A_s)\delta}$ (Cor 6.3) and Prop 6.3a shows the square-root scale is necessary along an explicit family for each fixed s . No growing s is substituted.

- **Uniform/sublinear:** Prop 6.4 threshold independent of s ; Thm 6.5 via SublinearEdit/[22] and RootScore; E35 uniform range $\forall s \leq S$ at $n \geq T(P(S))$. Checked statically.
- **Selected Theorem C chain:** `DefectExplicitPublication.rooted_defect_eventual := (E33.Fexp E34.NeditE s, E34.theoremC_fully_explicit_final)`; E34 chain uses `lemma11_V` (F.5). Alon–Shapira exclusion on the six public cones is a dynamic check (E4).
- **B7 / tower:** (3.11) link and (C.1)-(C.3) are formally stated; tower propagation not humanly rederivable from the text (A2-4). Inverse threshold: `sMaxT` requires $n \geq T(P(0))$ (stated).
- **gainOf conventions:** `LossBudget.budget_iff` and `SplitMixedGap.mixed_gap_zero` use `FarRounding.gainOf` ($C(|K|, 2)^{-1}$); partition-stability arguments use edge/defect accounting. Matches §7.2.

Gate E2 first-pass verdict

INCONCLUSIVE — no mathematical error found; all headline assemblies rederived; but four A.2 items (A2-3, A2-4, A2-5, A2-6) are essential or quantitative steps whose human-readable derivation is absent or only sketched, and one of them (A2-3) is stated in a way that does not by itself give the claimed feasibility. Their correctness therefore rests on the Lean development (to be confirmed in E4) and not on the manuscript. These are exposition findings, not counterexamples.

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